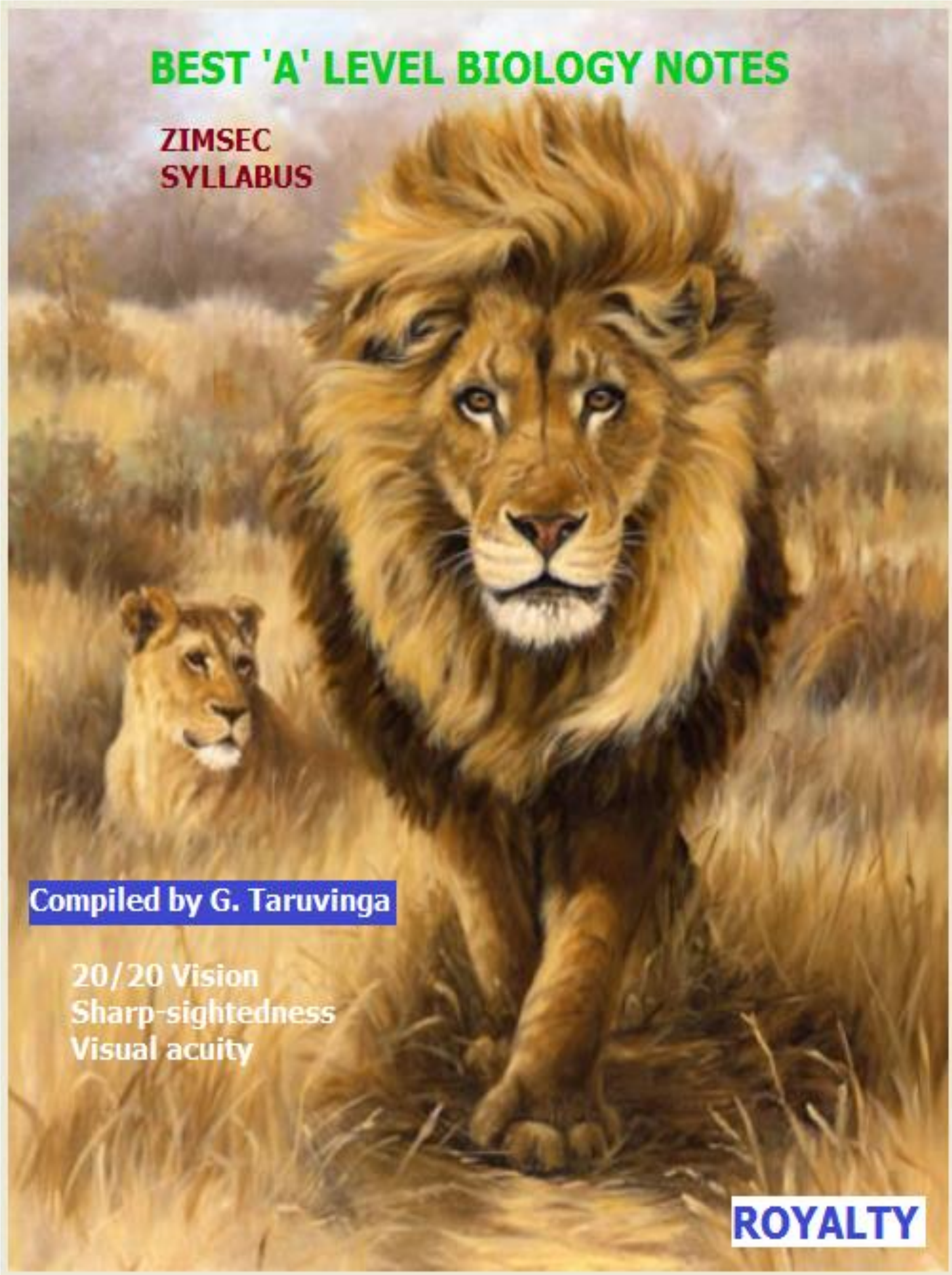

BEST 'A' LEVEL BIOLOGY NOTES
ZIMSEC
SYLLABUS
Compiled by G. Taruvinga
20/20 Vision
Sharp-sightedness
Visual acuity
ROYALTY

8.3 CELL AND NUCLEAR DIVISION

THE CELL CYCLE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.3.1 The Cell Cycle	1. outline the cell cycle 2. describe interphase	- Interphase - Mitosis - Cytokinesis - Growth - DNA replication	• Illustrating the cell cycle • Outlining DNA replication.	• ICT tools • Braille software/Jaws • Print media
8.3.2 Mitosis	1. describe the behaviour of chromosomes, nuclear envelope, cell membrane, centrioles and spindles during mitosis	- Prophase - Metaphase - Anaphase - Telophase	• Observing behavior of chromosomes in a root tip squash • Drawing of diagrams showing phases of mitosis.	
	2. distinguish between cytokinesis in plants and animals	- Cytokinesis	• Discussing cytokinesis in plant and animal cells.	
	3. explain the importance of mitosis	-Growth -Repair -Asexual reproduction -Production of genetically identical cells	• Discussing the importance of mitosis.	
	4. identify factors that increase chances of cancerous growth	- Carcinogens - Mutations - Radiation	• Discussing factors associated with cancerous growth.	
	5. outline the stages involved in the development of cancer	- Uncontrolled cell division	• Watching video clips. • Analysing video clips.	

- **CELL CYCLE** is the process of **cell growth** and **cell division. It** has three main phases: **interphase**, **mitosis**, and **cytokinesis**.
- **Cell growth** has **one phase** called **interphase** which has **three sub-phases** namely **G_1**, **S** and **G_2**.
  - **Interphase** is a period of synthesis and growth. The cell produces many materials required for its own growth and for carrying out all functions. DNA replication occurs during interphase.
- **Cell division** consists of **two phases**, **mitosis** and **cytokinesis**.
  - **Mitosis** (a.k.a. M-phase) is the **division of the nucleus** (nuclear division). It has **four sub-phases** namely **prophase**, **metaphase**, **anaphase** and **telophase**.

- **Cytokinesis** (a.k.a. C-phase) is the **division** of the **cytoplasm** into **two daughter cells** which usually, but not always, occurs immediately after nuclear division.
- Mitosis only takes up about 5 - 10% of the cell cycle. The majority of the cell cycle is taken up by Interphase

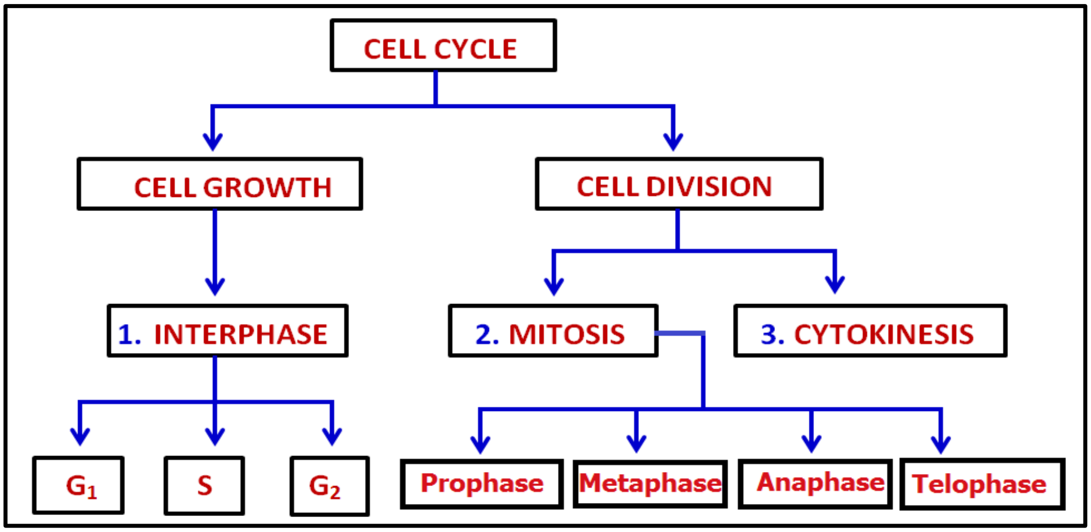


DIAGRAM: The phases of the cell cycle.

❖ The length of the cell cycle depends on:

1. the type of cell
2. external factors such as temperature, food and oxygen supplies.

– Bacteria may divide every 20 minutes, growing yeast cells divide every 30 minutes, cells epithelial cells of the intestine wall divide every 8-10 hours, animal sperm-producing cells every 18 to 24 hours, onion root-tip cells may take 20 hours to divide whilst many cells of the nervous system never divide at all.

– Fully differentiated cells generally remain arrested in the G1 stage and will not normally divide again. Some cells remain arrested in the G2 stage, for example, human cardiac muscle cells.

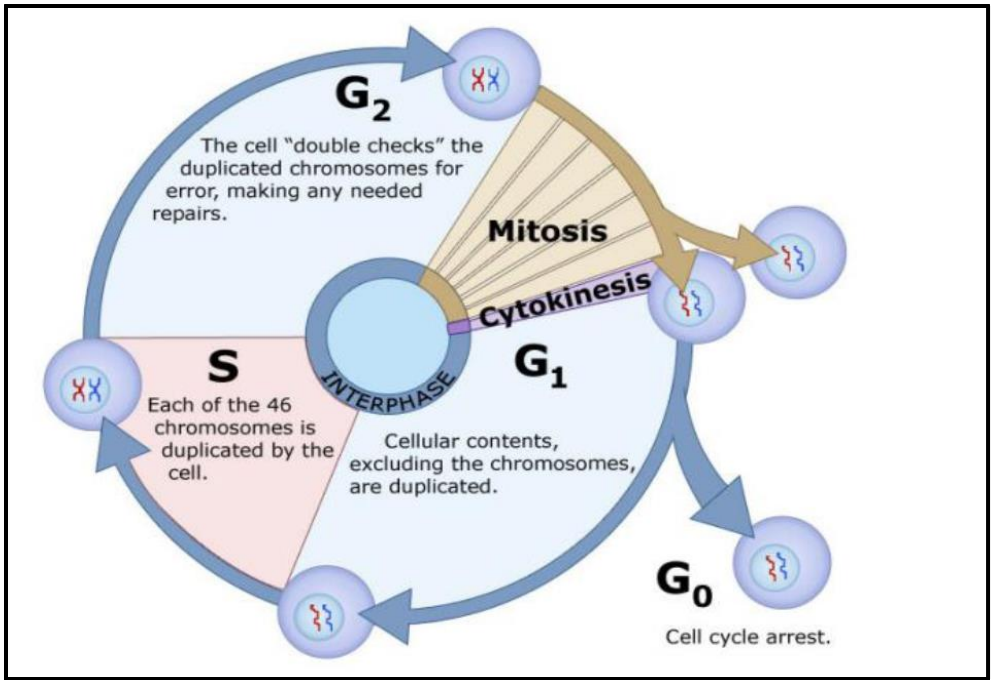
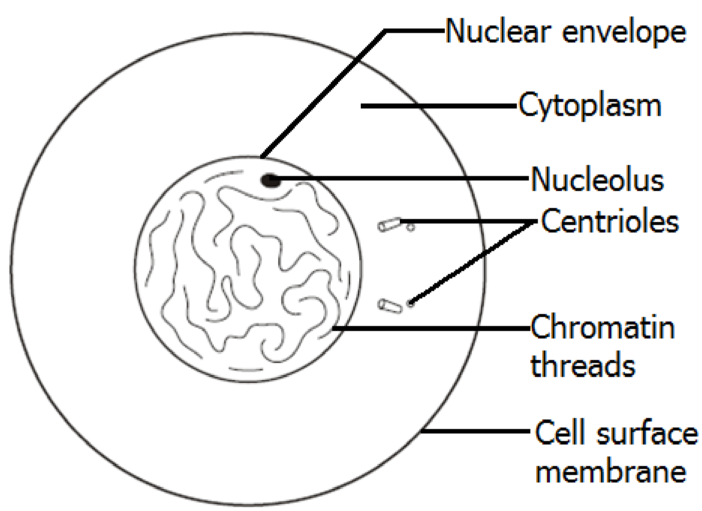


DIAGRAM: Phases of the cell cycle

– **The stages of the cell cycle are explained fully below.**

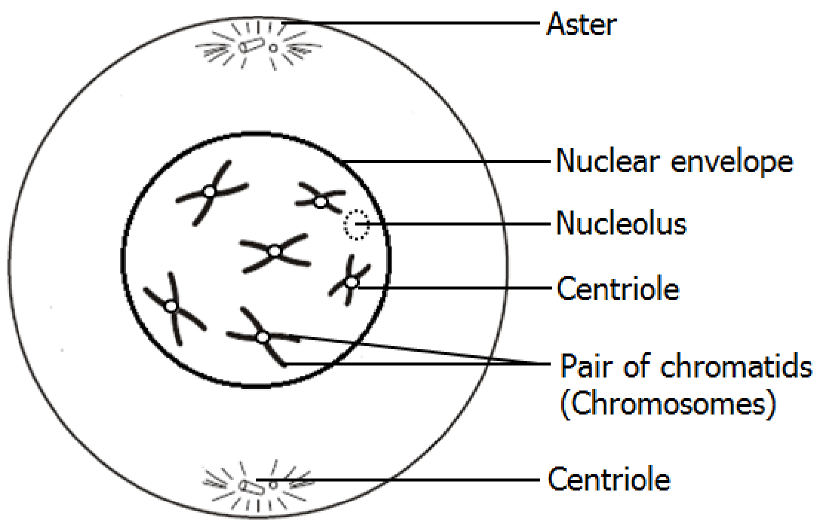
<table>
<tr><td colspan="2"><u>INTERPHASE</u>
– Interphase consists of three sub-phases:</td></tr>
<tr><td>1) G_1
▪ organelles are duplicated
▪ centrioles replicate
▪ Normal life processes:
• Respiraton
• Protein synthesis</td><td rowspan="3">

Interphase</td></tr>
<tr><td>2) S
▪ Chromosomes are duplicated/replication of DNA
▪ Semi-conservative replication of DNA</td></tr>
<tr><td>3) G_2
▪ Duplicated chromosomes are checked for any errors that may have occurred when copying them.</td></tr>
</table>

MITOSIS

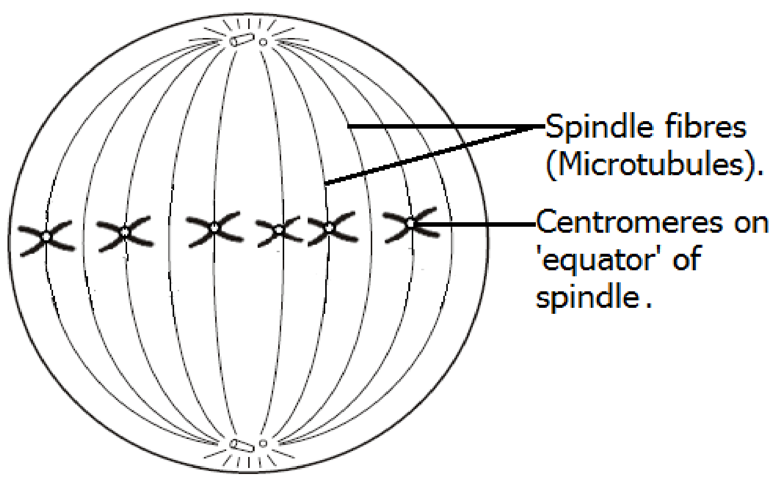
- Mitosis is the division of the nucleus (nuclear division) that leads to the formation of two identical daughter cells.
- Mitosis is also called **karyokinesis**. [The division of the **nucleus** is called **karyokinesis** as opposed to **cytokinesis** which is the division of the **cytoplasm**].
- Mitosis is **copying division** where chromosomes are **copied accurately** and passed on when cells divide to give **identical cells**.
- Mitosis consists of four sub-phases:

1. **Prophase** - Replicated chromosomes supercoil (shorten and thicken)
   - Chromosomes shorten and thicken (supercoil),
   - Chromosomes become visible
   - Chromosomes consist of two chromatids
   - Chromatids are joined by a centromere (central body).
   - Centrioles migrate to the poles of the cell (Not in Plant cells)
   - Spindle fibres form
   - Nuclear envelope breaks down

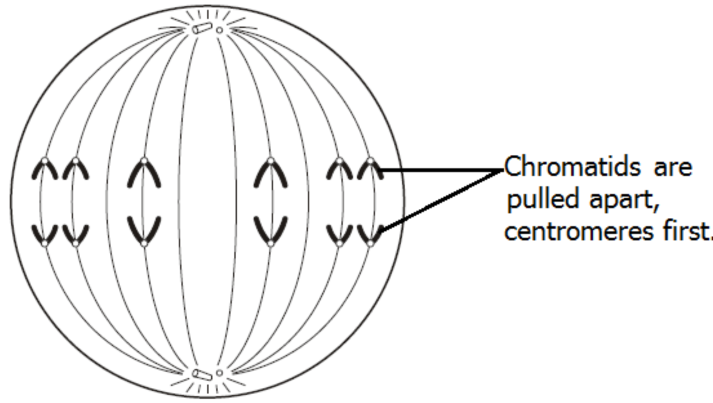
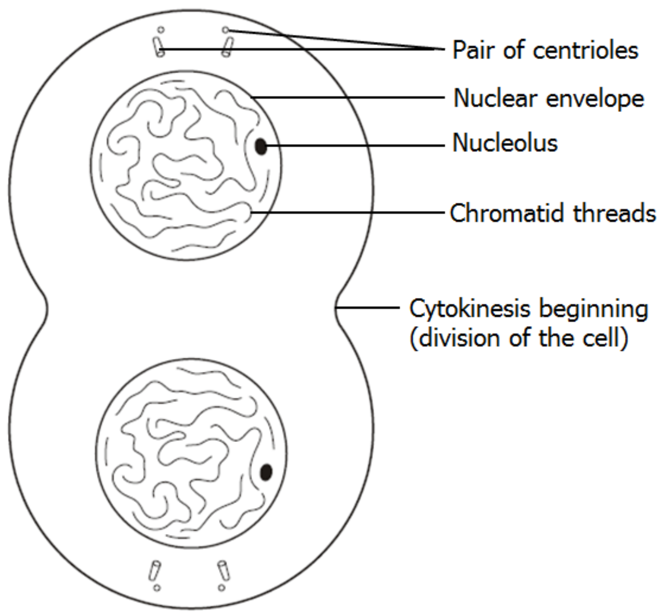


Prophase

2. **Metaphase** - Replicated chromosomes line up along the cells equator
   - Homologous pairs of chromosomes arrange themselves in a line along the cell's equator.
   - Spindle fibres attach to the centromere which holds the homologous pair together.



Metaphase

3. **Anaphase** - The replicas of each chromosome are separated from each other and pulled towards the poles of the cells ▪ Centromeres divide ▪ Separating each pair of chromatids ▪ The spindles contract and shorten ▪ Pulling chromatids to opposite ends of the cell, ▪ Centromere first.	 Anaphase
4. **Telophase** - Two new nuclei are formed ▪ The chromatids reach each pole of the cell. ▪ The chromatids uncoil and become long and thin again ▪ They are now referred to chromosomes again. ▪ A new nuclear envelope forms around each group of chromosomes so they are now two nuclei.	 Telophase

CYTOKINESIS

– Cytokinesis is cell division but specifically division of the cytoplasm to give two daughter cells. It occurs shortly after mitosis/karyokinesis.
  - The cytoplasm divides and the plasma membrane nips in half forming two new cells.
  - In animal cells this occurs by infolding of the plasma membrane, followed by formation of a cleavage furrow along the cell equator. The cleavage furrow deepens and eventually cuts the cell in two.
  - In plants this starts with the formation of a **phragmoplast** or **cell plate** which is laid down along the cell equator. A new membrane and cell wall is laid down along this cell plate (phragmoplast).

- The two new genetically identical cells formed then pass into the G_1 phase of the cell cycle.

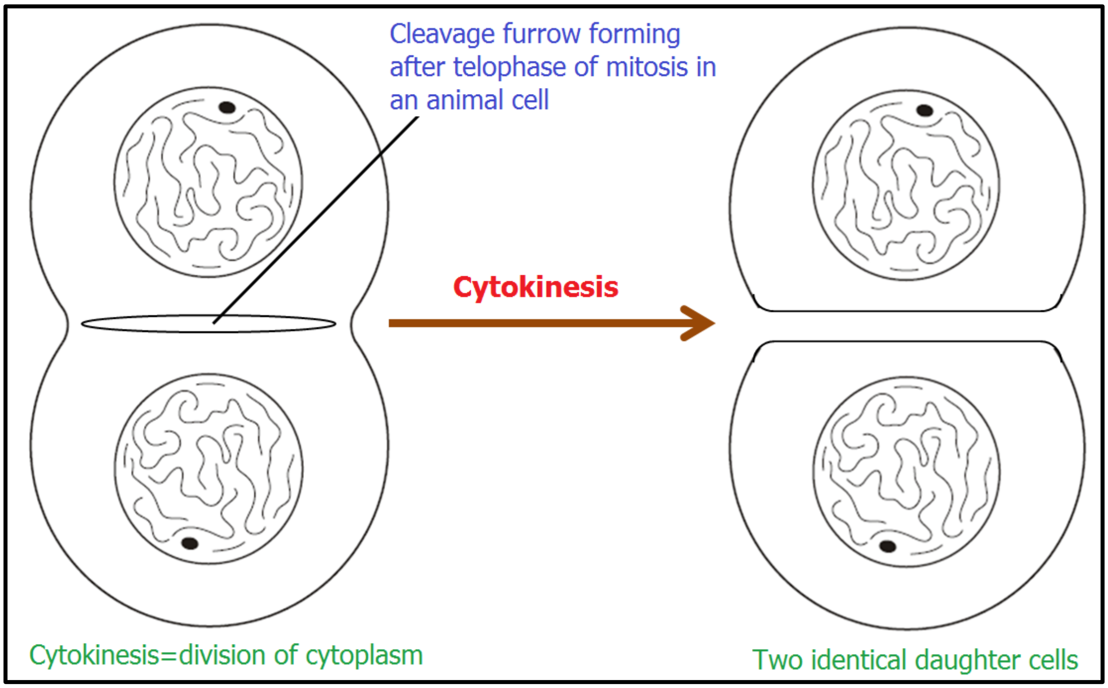


DIAGRAM: Cytokinesis in an animal cell.

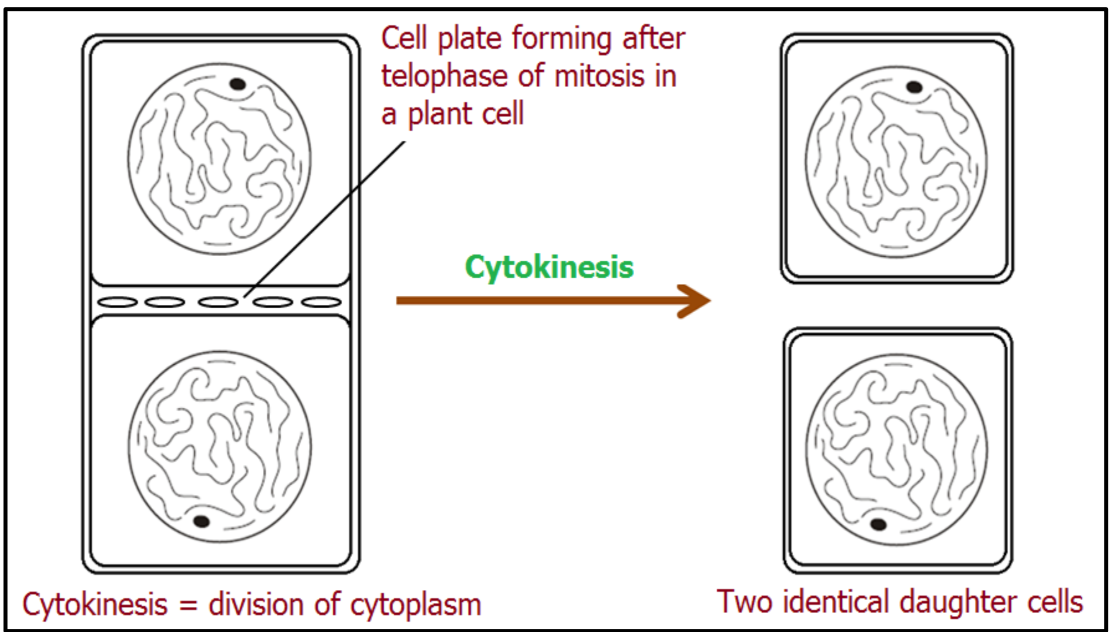


DIAGRAM: Cytokinesis in a plant cell.

❖ **The two new daughter cells formed are;**

- Diploid 2n (46 chromosomes as 23 of homologous pairs in humans).
- Genetically identical to each other and the parent cell.

- Most animal cells are capable of undergoing mitosis and cytokinesis.
- In plants only meristem cells can divide in this way.

❖ **What are homologous chromosomes?**

- A pair of identical chromosomes (1 from father and 1 from mother)
- Same genes but can have either the same or different alleles
- Humans have 23 homologous pairs of chromosomes

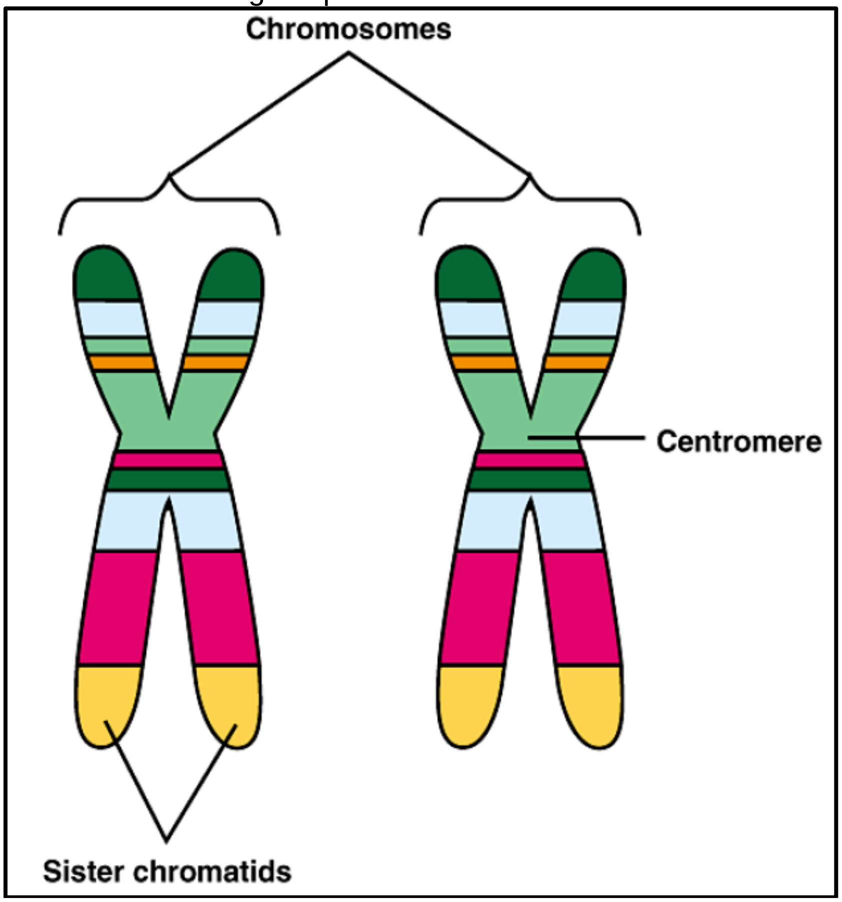


DIAGRAM: A pair of homologous chromosomes

THE SIGNIFICANCE OF MITOSIS

Explain the significance of mitosis for growth, repair and asexual reproduction in plants and animals.

1) Mitosis is type of cell division involving **somatic** or normal body cells.
2) It is important that the cells divide and **replace old worn out cells/repair tissues** and more importantly be able to replicate the duties of the cells they replace.
3) Mitosis is important for **growth of tissues to take place**. Mitotic divisions enable a single cell to grow e.g. from conception, repeated cell division has allowed us to developed into multicellular organisms.

4) It is also important for **genetic stability/maintains chromosome number (ploidy)**. By duplicating the exact copy of our genetic material it ensures that our genetic material is stable and able to carry out its function correctly because the instructions from the previous cells would have been passed on to the new daughter cells.
5) It's also important in **asexual reproduction** e.g. in plants and yeast cells the offspring produced are identical to the parent since they are produced as a result of mitotic cell division. As a consequence, the offspring have all the advantages of the parents in mastering the same habitat – and any disadvantages, too
6) Produces **genetically identical cells/clones.**

CANCER

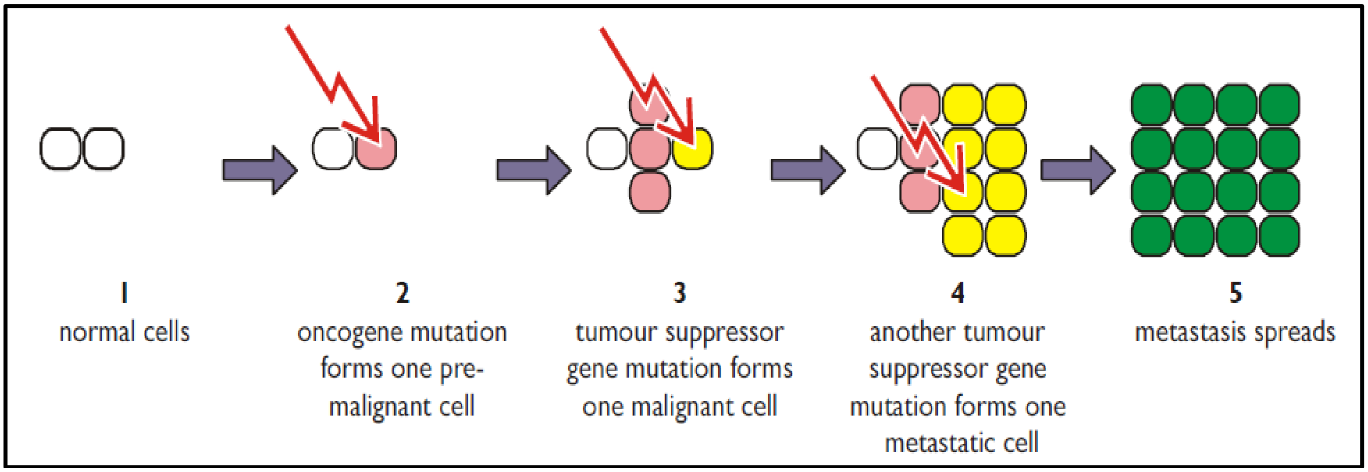
- The process of development of cancer is called **oncogenic transformation.**
- Normal cells have the property of contact inhibition (stoppage of growth on coming in contact with other cells), but cancer cells lose this property.
- Cell division in normal healthy cells is controlled by factors such as **cell cycle regulators** and **external chemical signals**. Cancerous cells do not respond to these **regulatory signals**.
- As a result, cancer cells divide uncontrollably to give rise to mass of cells called **tumours**.
- **Tumours are of 2 types** – benign and malignant.

1. Benign tumours – Remain confined to their original location and do not spread.
  - Most benign tumours (e.g. warts – see picture below) do not cause problems and can be successfully removed.

PICTURE: Warts are an example of a benign tumour.

2. Malignant tumours– These exhibit **metastasis** i.e., the cells that break off from such primary tumours spread via the circulatory system and lymphatic system and wherever they reach, new tumours called secondary tumours are formed.
  - Malignant tumours actually represent cancer. The cells actively divide, grow, and starve the normal cells of vital nutrients. Metastasis is what usually kills the patient.
  - Cancers become more common as one gets older.

- **Cancer cells contain mutated genes known as oncogenes.** The mutations are found in genes that are involved in mitosis; that is, in **genes that control the cell cycle**. These genes are **proto-oncogenes** and **tumour suppressor genes**.
- **Proto-oncogenes** encode proteins that act as **growth factors** by stimulating **cell division**. The proteins allow cells to pass the checkpoints in the cell cycle and are part of the normal tight control of cell division in organisms. Normally the proto-oncogene proteins are only expressed when needed for growth, but if a mutation causes a proto-oncogene to be expressed all the time, then it causes uncontrolled cell division, which can lead to cancer. **The mutated gene is then called an oncogene.**
- **Tumour-suppressor genes** encode proteins that **inhibit cell division** by blocking the cell cycle checkpoints. Some tumour suppressor genes also initiate cell death (apoptosis). **Tumour suppressor genes** thus **override** the effect of **oncogenes**, and so are also known as **anti-oncogenes**. For a cancer to occur, one or more tumour suppressor genes must also be mutated, so their control is removed and cells divide out of control.



- So a cancer is usually the result of several different mutations building up.

Steps in the development of cancer

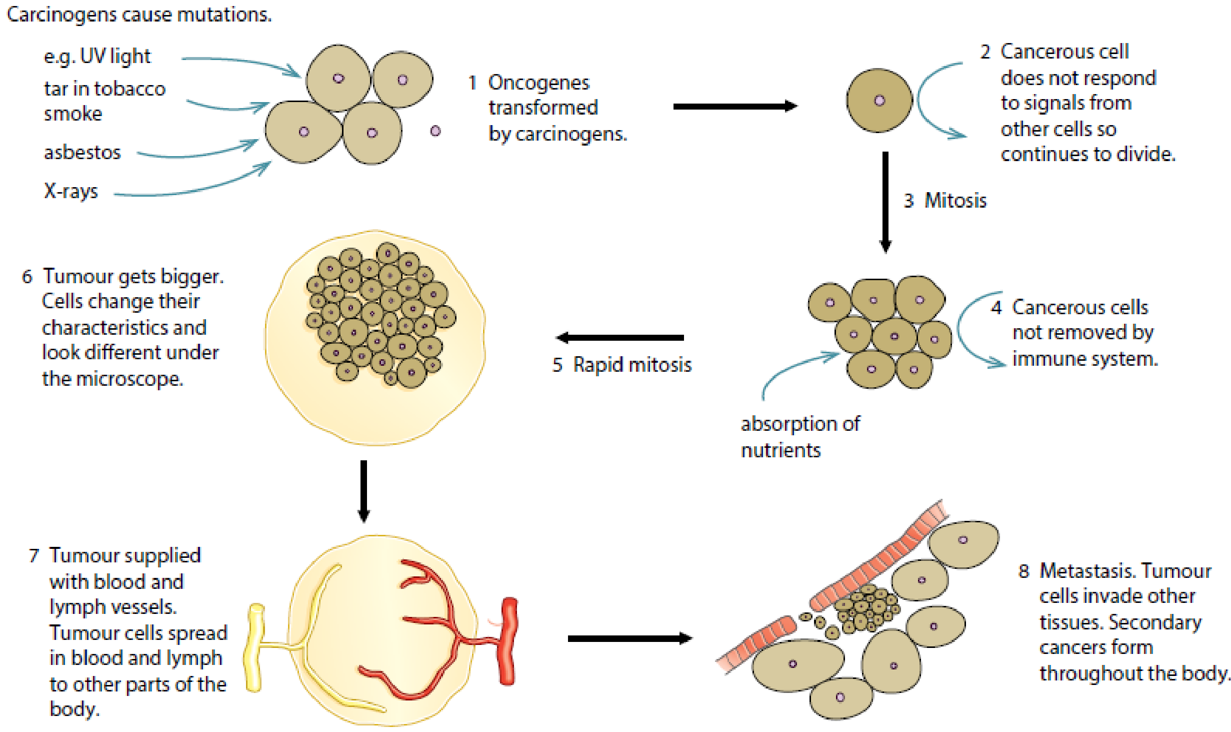
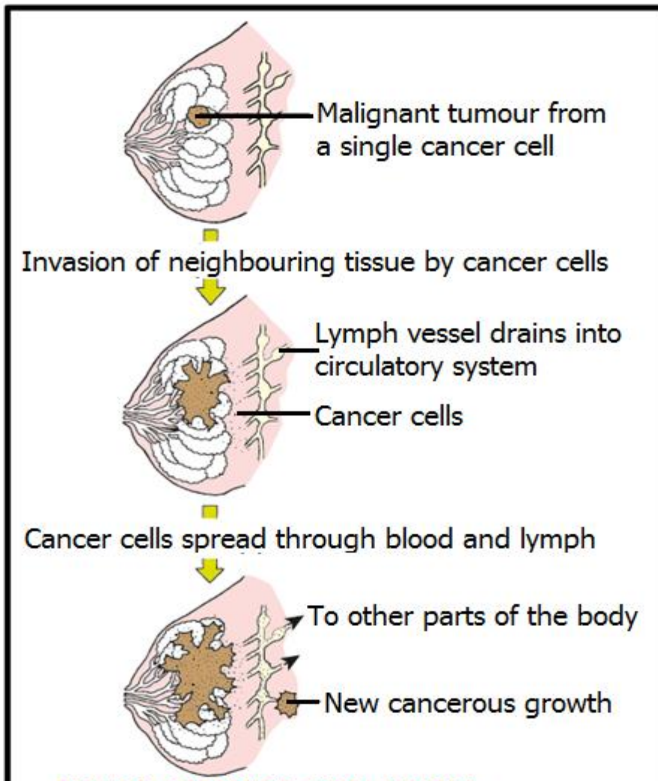
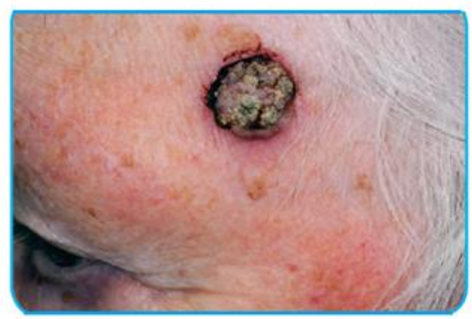


DIAGRAM: Stages in the development of cancer



BREAST CANCER SPREADING



PICTURE: Skin cancer

Metastasis is the spread of cancer to other parts of the body through the circlatory and lymphatic systems.The diagram on the left is showing the metastasis of breast cancer.

- **Causes of cancer**
  - **Carcinogens –** Physical, chemical, and biological agents that cause cancer Example - ionizing radiations (X-rays and radioactive sources emitting α, β and gamma radiation), non-ionizing radiations (UV), tar in tobacco smoke, asbestos dust.
  - **Oncogenic (cancer-causing) viruses** – they have viral oncogenes (cancer-causing genes) e.g. human papilloma virus (HPV) which causes cervical cancer.
  - **Mutations** of **proto-oncogenes** and **tumour suppressor genes:** Sometimes, due to mutations, normal genes in our body called **proto-oncogenes** an**d tumour suppressor genes** get **converted** into **cellular oncogenes** that cause cancer.
  - **Hereditary predisposition-**it runs in some families due to inherited genes. Individuals can inherit oncogenes or mutant alleles of tumor-suppressor genes e.g. in breast cancer.

- **Diagnosing cancer**
  - Biopsy and histopathological studies
  - **Biopsy** – Suspected tissue is cut into thin sections and examined microscopically
  - **Radiography**, CT scan (computed tomography), and MRI (Magnetic resonance imaging) are techniques of diagnosing cancers.
  - **C T Scan** – 3-D imaging of internals of an organ is generated by X-rays.
  - **MRI Scan** – Pathological and physiological changes in a living tissue are detected by using magnetic fields and non-ionising radiations.
  - Immunological and molecular biological diagnostic techniques can all be used to detect cancers.
  - Identifying certain genes, which make an individual susceptible to cancers, can help to prevent cancers.
- **Treatment of cancer**
  - **Radiotherapy** – Tumour cells are irradiated to death. Also, proper care is taken for protecting surrounding normal tissues.
  - **Chemotherapy** – Drugs specific for particular tumours are used to kill cancer cells. They have side effects such as hair loss, anaemia, etc.
  - **Immunotherapy**– Biological response modifiers such as α- interferons are used. They activate the immune system of patient and helps in destroying the tumour.

MEIOSIS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.3.3 Meiosis	1. explain the meanings of the terms haploid, diploid and homologous chromosomes	- Haploid - Diploid - Homologous - Chromosomes	• Illustrating haploid cells, diploid cells and homologous chromosomes.	• ICT tools • Braille software/Jaws
	2. Describe the behaviour of chromosomes, nuclear envelope, cell membrane, centrioles and spindles during meiosis	- Interphase - Meiosis I - Meiosis II - Cytokinesis	• Observing behaviour of chromosomes during pollen grain formation • Drawing of diagrams showing phases of meiosis.	• Print media
	3. discuss the importance of meiosis	- Gamete formation - Genetic variation	• Discussing the importance of meiosis.	• Microscope • Prepared slides • Flowers
	4. compare and contrast mitosis and meiosis	- Similarities and differences between mitosis and meiosis	• Discussing the similarities and differences.	

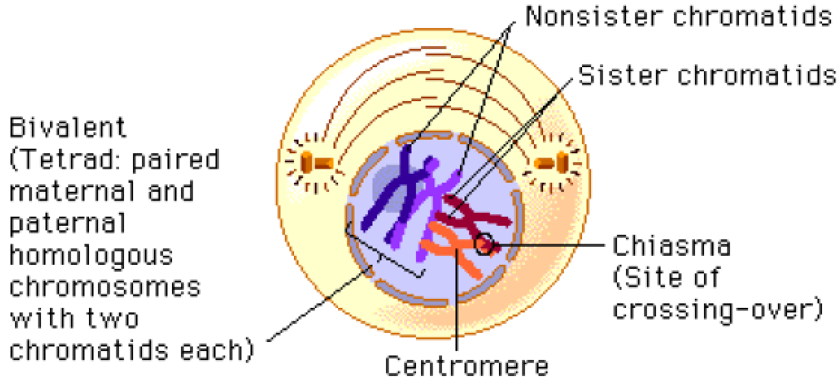
- ❖ The division of the nucleus of cells occurs in one of two ways:
- ❖ Mitosis produces 2 daughter nuclei with the same number of chromosomes as the parent cell and as each other. Occurs in somatic cells. Mitosis is copying division.
- ❖ Meiosis produces 4 daughter nuclei, each with half the number of chromosomes as the parent cell. Occurs in germline cells in gonads to produce gametes (sex cells).
- – Meiosis is a type of cell division known as a reduction division.
- – Produces gametes (Sex cells such as sperm, egg, pollen and plant egg cells)
- – Produces 4 haploid (n) cells from one parent diploid (2n) cell.
    - ▪ This is important as fusion of gametes nuclei at fertilization will need to result in a diploid (2n) cell.
    - ▪ Meiosis also mixes the genetic material before it is passed to the gametes so allowing for variation.

- – **Why is meiosis necessary?**
- – In sexual reproduction two gametes fuse to give rise to new offspring. The gametes must contain only one set (haploid number) of chromosomes to give a full set of chromosomes after fertilisation. If each gamete has a full set of chromosomes then the cell they produce has double the number.
- – This doubling of the number of chromosomes would continue at each generation.
- – In order to maintain a constant number of chromosomes in the adults the number must be halved during meiosis (reduction division).

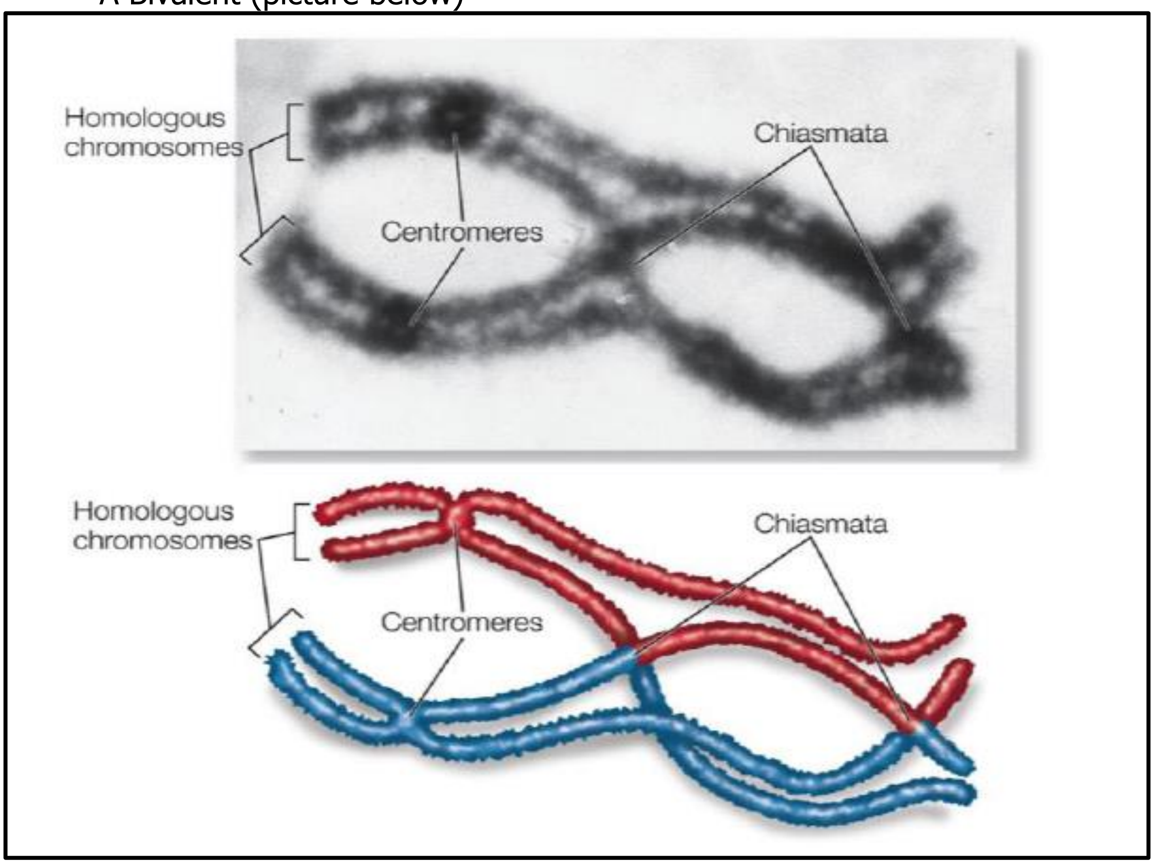
– Meiosis has two main stages: Meiosis I and Meiosis II, each with several sub-phases.

Meiosis I

❖ Prophase I



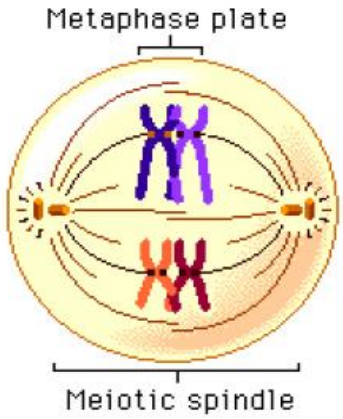
- Chromatin condenses and undergoes super-coiling, resulting in it becoming shorter, thicker and visible under a light microscope.
- The chromosomes now come together to form homologous pairs (bivalents)
- At this point, each homologous chromosome pair is visible as a bivalent, a tight grouping of two chromosomes, each consisting of two sister chromatids.
- A Bivalent (picture below)



- The sites of crossing-over are seen as crisscrossed non-sister chromatids and are called chiasmata (singular: chiasma).

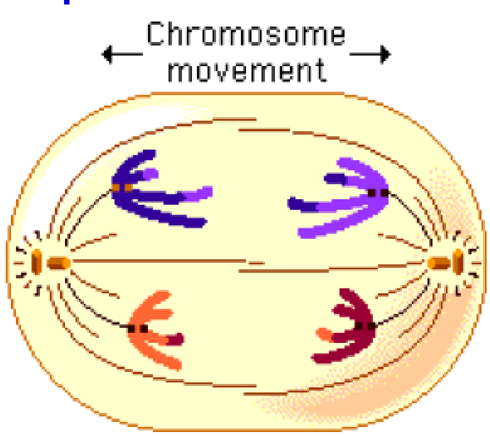
- These allow swapping of paternal and maternal alleles.
- The nuclear envelope and nucleolus break down.
- Centrioles and spindle fibres form (no centrioles in plant cells).

❖ Metaphase I



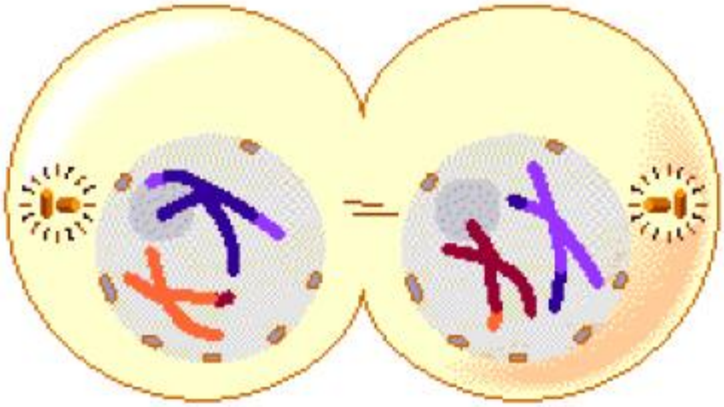
- Centrioles (not in plant cells) migrate to the poles of the cell
- Bivalents are arranged on a plane equidistant from the poles called the metaphase plate or equator
- Independent random assortment of Bivalents, where the orientation of the bivalent on the metaphase plate is random and independent of the orientation of the bivalents either side of it.
- Spindle fibres attach to the centromeres of sister chromatids.

❖ Anaphase I



- Two chromosomes of each bivalent separate and start moving toward opposite poles of the cell as a result of the action of the spindle.
- The sister chromatids remain attached at their centromeres and move together toward the poles.
- A key difference between mitosis and meiosis is that sister chromatids remain joined after metaphase in meiosis I, whereas in mitosis they separate (centromeres break)
- At this point recombinants are formed from the crossing over events of prophase I.

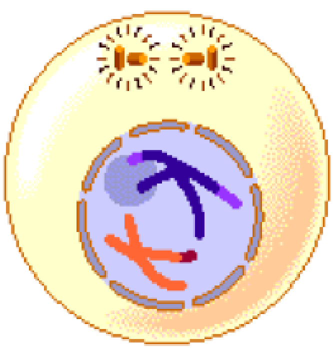
- **Telophase I** (Not in plant cells)

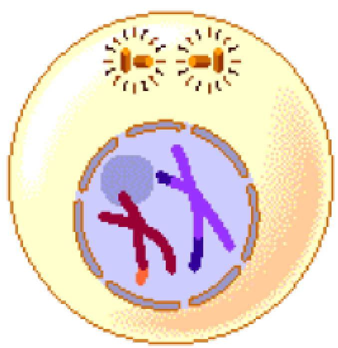


- The homologous chromosome pairs complete their migration to the two poles as a result of the action of the spindle.
- A nuclear envelope reforms around each chromosome set, the spindle disappears, and cytokinesis follows.
- In animal cells, cytokinesis involves the formation of a cleavage furrow, resulting in the pinching of the cell into two cells.

Meiosis II

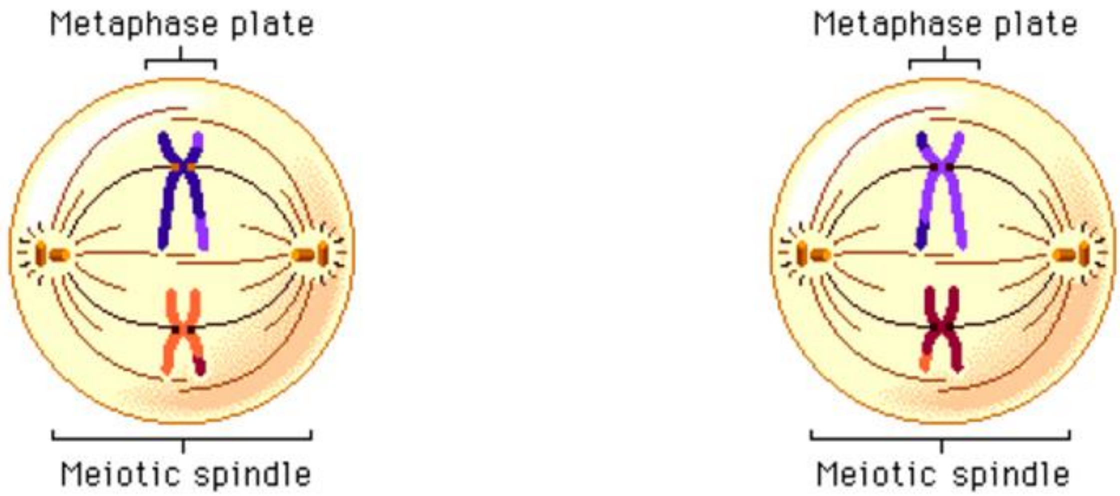
- **Prophase II**





- The centrioles duplicate. This occurs by separation of the two members of the pair, and then the formation of a daughter centriole perpendicular to each original centriole.
- The nuclear envelope breaks down, and the spindle apparatus forms.

❖ Metaphase II

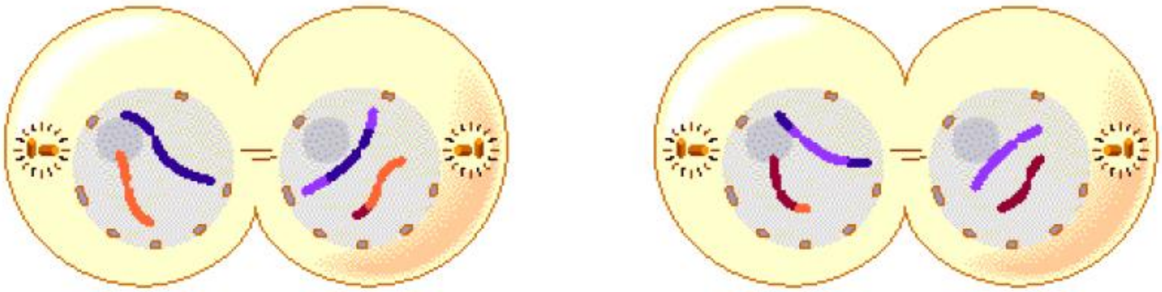


- Each of the daughter cells completes the formation of a spindle apparatus.
- Single chromosomes align on the metaphase plate, much as chromosomes do in mitosis. This is in contrast to metaphase I, in which homologous pairs of chromosomes align on the metaphase plate.
- Independent random assortment of chromatids, where the orientation of the recombinant chromatids on the metaphase plate is random and independent of the orientation of the recombinant chromatids either side of it.
- Spindle fibres attach to the centromeres of recombinant chromatids

❖ Anaphase II

- The centromeres separate, and the two chromatids of each chromosome move to opposite poles on the spindle.
- The separated chromatids are now called chromosomes in their own right.

❖ Telophase II



Four haploid daughter cells

- A nuclear envelope forms around each set of chromosomes.
- Cytokinesis takes place, producing four daughter cells (gametes, in animals), each with a haploid set of chromosomes.
- Because of crossing-over, some chromosomes are seen to have recombined segments of the original parental chromosomes.

Why is Meiosis Important?

1. Chromosome number is halved- prevents chromosome number from doubling each generation
2. Meiosis ensure genetic variation
   (a) Crossing over: exchange of genetic info ensures gametes carry different gene combos therefore different offspring from same parent
   (b) Segments of chromatids of homologous chromosomes are exchanged
   (c) Chiasmata develop at a point where 2 pairs cross- mixing of genes
   (d) Metaphase 1: homologous chromosome pairs are arranged randomly on equator Random Assortment= variation of genes, the Law Of Independent Assortment
   (e) Metaphase 2: chromatids line up randomly of equator- random assortment
3. Therefore, crossing over, metaphase 1 and 2 = genetic variation.

Explain how meiosis and fertilisation can lead to variation through the independent assortment of alleles.

- Meiosis and fertilisation create huge levels of variation in individuals of the next generation.
- This results from 6 major events:

❖ **Crossing Over**

- Prophase I
- Homologous pairs of Chromosomes (Bivalents) form
- Chiasma(ta) form between homologous chromosomes
- Alleles swap between paternal and maternal chromatids.
- Maybe 3 or 4 crossing over events per Bivalent

❖ **Independent random assortment of Bivalents**

- Metaphase I
- Bivalents line up along the cell equator randomly and independent of each other

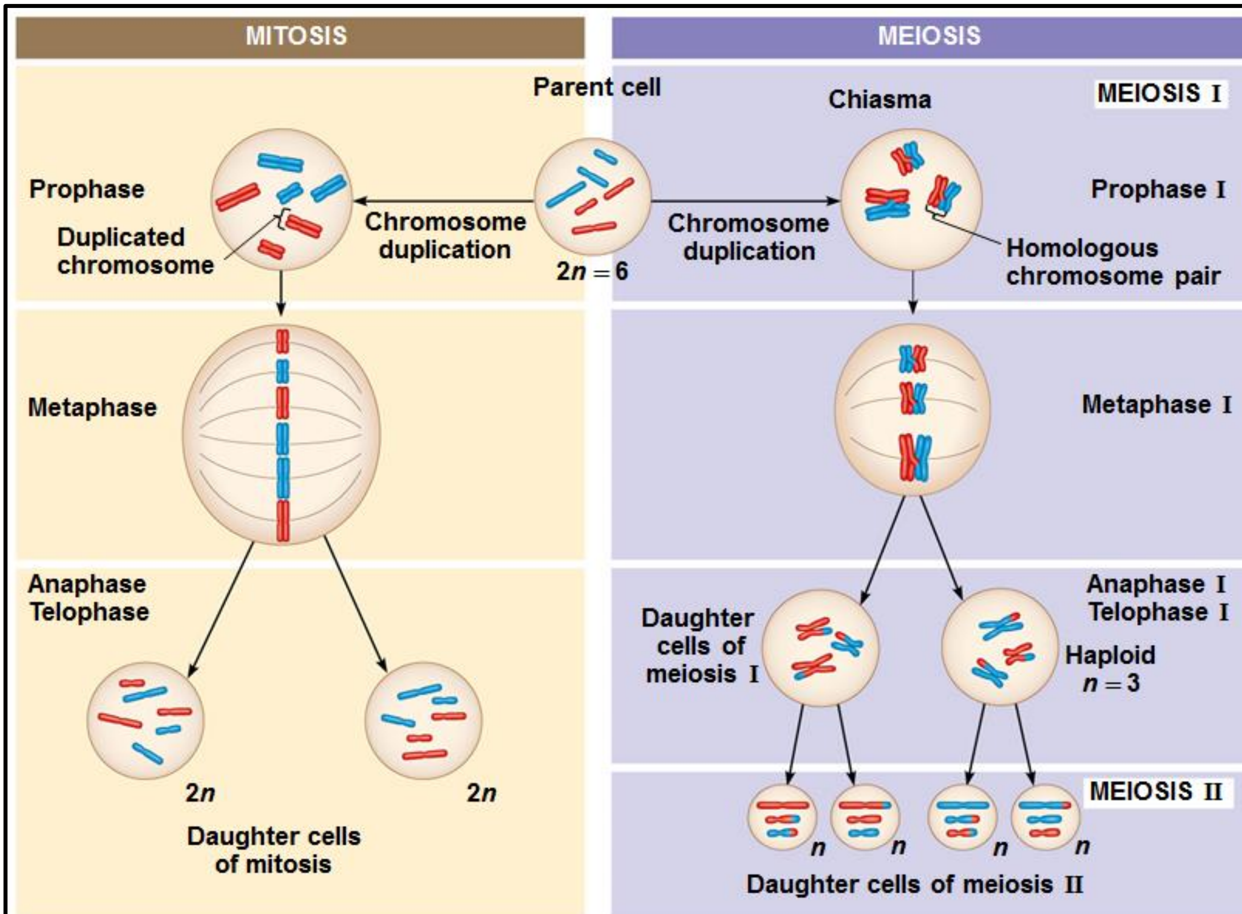
- **Independent random assortment of (recombinant) chromatids**
  - Metaphase II
  - Chromatids line up along the cell equator randomly and independent of each other.
- **Random mating**
  - Choices of mates is random
- **Random fusion of gametes**
  - Chances of gamete fertilising another specific gamete
- **Chromosome mutations**
  - Any changes to DNA further increases the variation

Differences between mitosis and meiosis

- Mitosis conserves the number of chromosome sets, producing cells that are genetically identical to the parent cell
- Meiosis reduces the number of chromosomes sets from two (diploid) to one (haploid), producing cells that differ genetically from each other and from the parent cell

MITOSIS	MEIOSIS
• Occurs in somatic cells - Development of zygote to embryo - Growing regions (bone, skin, damaged area) • Produce two identical daughter cells • Needed for growth/ repair • I-P-M-A-T Differences: • One cell division • Two daughter cells same number chromosomes • Somatic cells formed genetically similar to parent cell	• Occurs in reproductive organs • Testes/ ovaries (animal) anthers/ovaries (plant) • Occurs only once • Diploid to haploid (half number of chromosomes-prevent doubling up) • Produces gametes(animal) or spore (plant) • Enables fertilization to occur • Produces four daughter cells that have genetic variation • Has 2 stages - 1= reduction to 2 haploid cells. 2= reduction to 4 haploid cells • Two cell divisions • Four daughter cells formed-haploid • Gametes are formed genetically different to parent cell

Property	Mitosis	Meiosis
DNA replication	Occurs during interphase before mitosis begins	Occurs during interphase before meiosis I begins
Number of divisions	One, including prophase, metaphase, anaphase, and telophase	Two, each including prophase, metaphase, anaphase, and telophase
Synapsis of homologous chromosomes	Does not occur	Occurs during prophase I along with crossing over between nonsister chromatids; resulting chiasmata hold pairs together due to sister chromatid cohesion
Number of daughter cells and genetic composition	Two, each diploid (2*n*) and genetically identical to the parent cell	Four, each haploid (*n*), containing half as many chromosomes as the parent cell; genetically different from the parent cell and from each other
Role in the animal body	Enables multicellular adult to arise from zygote; produces cells for growth, repair, and, in some species, asexual reproduction	Produces gametes; reduces number of chromosomes by half and introduces genetic variability among the gametes



A comparison of mitosis and meiosis in diploid cells

PRACTICE QUESTIONS

1. Explain the significance of mitosis.

- It brings about the growth of an organism:
- It brings about asexual reproduction.
- Ensures that the chromosome number is retained/ensures genetic stability.
- Ensures that the chromosomal constitution of the offspring is the same as the parents /production of clones.
- Replacement of old worn out cells/repair tissues.

2. Explain the significance of meiosis.

- Formation of gametes that contain half the number of chromosomes as the parent cells.
- Restoration of the diploid chromosomal constitution in a species at fertilisation.
- It brings about new gene combinations that lead to genetic variation in the offspring.

3. Although mitosis is a continuous process, for ease of reference it is conventionally divided into the following stages: interphase, prophase, metaphase, anaphase and telophase.
   (a) Name the stages of mitosis during which,
      (i) the chromatids separate and move to the poles. [1]
      (ii) the nuclear membrane reforms and cytokinesis follows. [1]
      (iii) the chromosomes align on the equator of the spindle. [1]
      (iv) the chromosomes become stainable and the spindle starts to form. [1]
   (b) If the amount of DNA present in the cell at metaphase is 20 units, how much DNA will be present in each nucleus:
      (i) at the start of prophase. [1]
      (ii) immediately after telophase? [1] [Total:6]

3 (a) (i) anaphase; (ii) telophase; (iii) metaphase; (iv) prophase; 4
(b) (i) 20 units; (ii) 10 units; 2

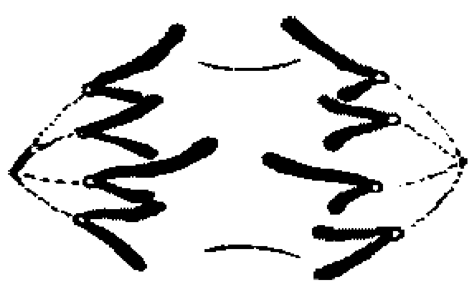
4. Read through the following passage about mitosis and then complete it by writing the most appropriate word or words in the spaces.

In flowering plants the process of mitosis is restricted to the apical .... and to .... In growing mammals mitosis can occur throughout the body.
However, not all regions of the young mammal grow at the same rate and this is called .... growth.

In the cell cycle, replication of DNA occurs in the..........................phase, after which there is a lag or gap phase, called the .... phase, before actual mitosis starts. The chromosomes also replicate before the onset of mitosis, but this replication is not visible until the middle of the .... stage.
At this stage, each chromosome consists of two ....held together by a ....
In the kangaroo, there are 10 pairs of chromosomes. Thus in mitosis an anaphase cell will contain .... chromosomes with .... migrating to each pole. The daughter cells therefore have the .... number of 2n = ... [12]

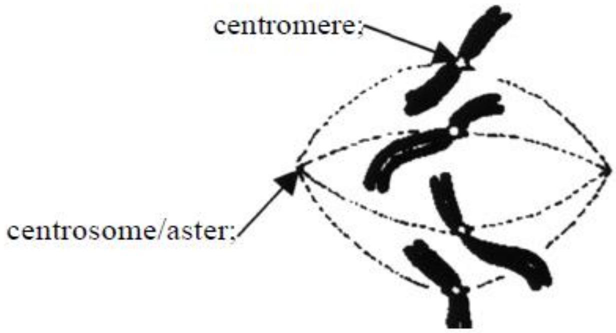
4. meristems; buds/intercalary meristems; allometric; S; G2; prophase; chromatids; centromere; 40/20 pairs; 20/10 pairs; diploid; 20; [12]

5. The drawing below shows a dividing animal cell nucleus at the anaphase stage of mitosis



(a) Briefly describe what you can see happening in this stage of mitosis. [4]
(b) (i) Draw accurately the appearance of the same nucleus at the metaphase stage of mitosis. [3]
(ii) On your drawing label a centromere and a centrosome(aster). [2] [Total 9]

5 (a) spindles formed from centrosomes/centrioles; (daughter/replicated) chromosomes migrating to the poles; pulled by contracting spindles; which are attached to the centromeres; one set of chromosomes goes to one pole and other set to the other pole; max [4]
(b) (i) and (ii)
Drawing:
4 chromosomes not yet replicated;
attached to spindles by their centromeres;
same chromatid length/centromere positions as in anaphase drawing; [5]



6. (a) The statements in the list below the table describe some of the stages of mitosis. Complete the table by writing in the correct statement with the appropriate stage.
nuclear membranes reappear
DNA replicates
division of the cytoplasm occurs
chromosomes become shorter and thicker
chromosomes attach to spindle ends at equator
daughter chromosomes move to the poles [6]

Stage	Description
Prophase	
Metaphase	
Anaphase	
Telophase	
Interphase	
Cytokinesis	

(b) How does cytokinesis in plants differ from cytokinesis in animals? [2]
[Total 8]

6 (a)

Stage	Description
Prophase	chromosomes become shorter and thicker;
Metaphase	chromosomes attach to spindle ends at equator;
Anaphase	daughter chromosomes move to the poles ;
Telophase	nuclear membranes reappear;
Interphase	DNA replicates ;
Cytokinesis	division of the cytoplasm occurs;

(b) (in animals) cytoplasm divides by constriction (between daughter nuclei); (in plants) a phragmoplast/cell plate/new cell wall is synthesised (between the daughter nuclei); [2]